Task 1: E-Commerce Data Cleaning Report

Author: Kalyani Kurne
Dataset: Messy E-Commerce Sales Dataset 2026
Source: Kaggle Dataset

1. Project Overview

This project focuses on inspecting and cleaning a synthetic e-commerce dataset downloaded from Kaggle. The raw dataset contained 12,180 transaction records and 13 columns covering orders, customers, products, quantities, prices, discounts, payments, locations, order statuses, and ratings.

The objective was to identify data quality issues, apply justified corrections, and document each cleaning decision. Excel was used for visual inspection, while Python and Pandas were used for systematic checks and cleaning in Jupyter Notebook.

2. Data Quality Issues Identified

1. **Duplicate Records:** The raw dataset contained 180 extra exact duplicate rows, which could inflate transaction counts and sales calculations.

1. **Missing Data:** After duplicate removal, missing values were found in three columns:

Column	Missing Count	Missing Percentage
Discount_Percent	722	6.02%
Shipping_City	494	4.12%
Customer_Rating	6,056	50.47%

Percentages were calculated using the 12,000 records remaining after duplicate removal. All other original columns had no null values.

3. **Mixed Date Formats:** Order_Date was stored as text and contained five different date formats. No entries containing only a four-digit year were found.
4. **Non-Positive Quantities:** Quantity contained 96 negative values and 90 zero values, affecting 186 records, or 1.55% of the deduplicated dataset.
5. **Inconsistent Payment Labels:** Payment_Method contained 20 labels due to differences in spelling, capitalization, punctuation, and spacing.
6. **Inconsistent Country Labels:** Country contained 24 labels, including abbreviations, full names, and capitalization variations.

3. Cleaning Process

1. **Initial Inspection:** The dataset was inspected to understand its structure, missing values, duplicate records, unique labels, and numerical ranges.
2. **Duplicate Removal:** All columns were compared to identify exact duplicate rows. The 180 extra copies were removed while retaining one copy of each record. This reduced the dataset from 12,180 to 12,000 rows. Verification confirmed zero remaining exact duplicates.
3. **Missing Shipping Cities:** The 494 missing Shipping_City values were filled with “Unknown” because the actual city information was unavailable. No city names were guessed, and no rows were removed.
4. **Missing Customer Ratings:** Ratings were examined by Order_Status. All records with Cancelled, Pending, Returned, or Shipped statuses had missing ratings. Another 634 Delivered records had no recorded rating.

Missing ratings were retained because zero or mean imputation would introduce unsupported feedback values.

5. **Missing Discounts:** The 722 missing Discount_Percent values were retained. A missing discount was not assumed to mean zero discount. A Boolean column, Discount_Missing, was added to identify these records.
6. **Payment Standardization:** The 20 payment labels were standardized into six methods: Cash on Delivery, PayPal, Credit Card, Net Banking, UPI, and Debit Card.
7. **Country Standardization:** The 24 country labels were standardized into seven full names: United Kingdom, Germany, United Arab Emirates, United States, Australia, India, and Canada.
8. **Date Conversion:** Order_Date was converted from text to datetime using explicit parsing formats. All 12,000 dates were successfully parsed. Month-day-year was assumed for month-first hyphenated dates, and day-month-year for day-first slash-separated dates, based on the observed patterns.
9. **Quantity Flagging:** Non-positive quantities occurred across all order statuses, including Delivered orders. Therefore, negative quantities were not automatically treated as returns. Original values were retained, and an Invalid_Quantity flag was added to identify the 186 affected records.

4. Validation Checks

Check	Result
Exact duplicates after removal	0

Check	Result
Missing shipping cities after replacement	0 null values; 494 labelled “Unknown”
Unsuccessfully parsed dates	0
Missing, zero, or negative unit prices	0
Unit price data type	float64
Ratings outside 1–5	0 among non-missing values
Discounts outside 0–100	0 among non-missing values
Observed non-missing discounts	0, 5, 10, 15, 20, and 25
Invalid Order_ID formats	0; all matched ORD followed by six digits
Invalid Customer_ID formats	0; all matched CUST followed by five digits
Decimal quantities	None found
Category and order status labels	Eight categories and five statuses; no obvious label variations
Product names	46 unique names; no obvious spelling or capitalization variations
Leading/trailing spaces	None in Product_Category, Product_Name, or Order_Status

Check	Result
Product–category consistency	Each product appeared in only one category
City–country consistency	Each known city was associated with only one country

The city–country check confirmed consistency within the dataset; it did not independently verify geographical accuracy.

5. Additional Columns

Added Column	Purpose	Records Flagged
Discount_Missing	Identifies unavailable discount percentages	722
Invalid_Quantity	Identifies quantities less than or equal to zero	186

Both columns contain True/False values. They preserve the original data while making unresolved issues identifiable.

6. Tools and Libraries Used

- **Excel:** Visual inspection of the dataset.
- **Python:** Data validation and cleaning.
- **Pandas:** Missing-value analysis, duplicate removal, grouping, label standardization, and date conversion.
- **Numpy :** Used numerical calculations
- **Jupyter Notebook:** Code execution and documentation.

7. Final Results

Metric	Result
Original rows	12,180
Original columns	13
Exact duplicate rows removed	180
Final rows	12,000
Final columns	15
Remaining exact duplicates	0
Remaining missing discounts	722
Remaining missing ratings	6,056
Non-positive quantities flagged	186

Duplicates were removed, payment and country labels were standardized, and dates were converted to datetime. Missing cities were labelled “Unknown.” Unavailable discounts and ratings were retained, and non-positive quantities were flagged rather than replaced with guessed values.

8. Limitations and Assumptions

- The dataset is synthetic and does not represent actual customer transactions.
- Missing discounts remain unknown and must not automatically be treated as zero when calculating discounted amounts.
- Rating analysis should use available ratings and report the number of rated orders.

- The 186 non-positive quantity records require further clarification and will be excluded from sales quantity and revenue calculations.
- Ambiguous numeric dates were interpreted using the documented format assumptions.
- Total Amount was not present in the original dataset and has not been created during this cleaning stage.
- Final analysis must apply appropriate filters for order status and flagged records.

9. Output Files

The cleaned dataset is prepared for export as **ecommerce_cleaned.csv**.

Task 2: Sales Data Analysis

Project Overview

This analysis examines sales performance and customer purchasing patterns in the cleaned synthetic e-commerce dataset. Python and Pandas were used to calculate KPIs, summarize monthly revenue, and analyze repeat customers.

The cleaned dataset contains 12,000 records. Sales analysis uses 6,082 eligible delivered orders with positive quantities and available discounts.

Analysis Rules

Total Amount was calculated using:

Total Amount = Quantity × Unit Price × (1 − Discount Percentage ÷ 100)

Amounts were rounded to two decimal places and measured in USD. The discount was assumed to apply to the full product value. Tax and shipping charges were excluded.

Records with non-positive quantities or missing discounts were excluded from the sales subset. Of 6,578 delivered records, 6,082 were eligible and 496 were excluded. Therefore, revenue coverage was 92.46% of delivered records by count.

KPI Summary

KPI	Result
Revenue from Eligible Orders	USD 719,962.24
Total Eligible Orders	6,082
Average Order Value	USD 118.38
Total Units Sold	12,582
Top Category by Revenue	Electronics
Top Product by Revenue	Smartwatch

1. **Revenue:** The sum of discounted amounts for eligible orders was USD 719,962.24. This does not represent revenue from every delivered order.
2. **Total Eligible Orders:** There were 6,082 unique Order_ID values in the eligible sales subset.
3. **Average Order Value:** Eligible revenue divided by eligible orders produced an average value of USD 118.38 per order.
4. **Total Units Sold:** Eligible orders contained 12,582 units. This measure uses the same subset as the revenue KPI.
5. **Top Category:** Electronics generated the highest category revenue. This ranking measures revenue contribution rather than units sold.

6. **Top Product:** Smartwatch generated the highest product revenue.

Monthly Sales Trend

Monthly eligible revenue was analyzed from January 2024 to June 2026.

- **Highest month:** April 2024 — USD 28,232.78.
- **Lowest month:** November 2024 — USD 20,466.64.

Revenue fluctuated during the period without a consistent upward trend. Order volume, product mix, and discounts could be investigated to understand these differences. The analysis does not establish the causes of monthly changes.

Customer Insights

The eligible sales subset contained **2,883 unique customers**. Of these, **1,823 customers** placed more than one eligible order.

Repeat customers contributed **82.94% of eligible revenue**, showing that most revenue in this subset came from customers with multiple purchases.

Repeat-customer status was defined using eligible orders within the dataset. It does not represent the customer’s complete purchase history.

Key Findings

1. Eligible delivered orders generated USD 719,962.24 in revenue.
2. The eligible subset contained 6,082 orders and 12,582 units.
3. Average eligible order value was USD 118.38.

4. Electronics was the highest revenue-generating category.
5. Smartwatch was the highest revenue-generating product.
6. April 2024 had the highest monthly revenue, while November 2024 had the lowest.
7. Repeat customers contributed 82.94% of eligible revenue.
8. Payment-method order shares were similar, with PayPal slightly ahead at 17.05%.

Recommendations

- Explore customer retention initiatives because repeat customers contributed most eligible revenue.
- Review inventory availability and profitability for Electronics and Smartwatch before making stocking decisions.
- Investigate order volumes and product mix behind monthly revenue differences.
- Recover missing discounts and clarify non-positive quantities to improve analytical coverage.

Limitations

- The dataset is synthetic; findings demonstrate analytical methods rather than actual business performance.
- Sales KPIs exclude records with missing discounts or non-positive quantities.
- Revenue represents discounted product value and excludes tax and shipping.
- Delivered status is used as the completed-sales criterion; payment collection is not independently verified.
- Customer insights are based only on eligible orders.

- Ambiguous numeric dates were interpreted using the documented format assumptions.

Task 3: Visualization Challenge

Objective

The objective was to communicate sales findings using a bar chart, line chart, pie chart, and a static dashboard mockup. All visuals use the same eligible sales subset.

1. Bar Chart: Revenue by Category

Purpose: Compare revenue contributions across product categories.

Interpretation: Electronics generated the highest eligible revenue. The chart compares revenue rather than product demand or units sold.

Output File: category_revenue.png

Insert the category revenue chart here.

2. Line Chart: Monthly Revenue

Purpose: Show changes in eligible revenue over time.

Interpretation: April 2024 recorded the highest revenue at USD 28,232.78, while November 2024 recorded the lowest at USD 20,466.64. Revenue fluctuated without a consistent upward trend. The chart does not explain the causes of these changes.

Output File: monthly_revenue.png

Insert the monthly revenue chart here.

3. Pie Chart: Orders by Payment Method

Purpose: Compare each payment method’s share of eligible orders.

Payment Method	Eligible Orders	Share
Cash on Delivery	1,025	16.85%
Credit Card	1,032	16.97%
Debit Card	985	16.20%
Net Banking	993	16.33%
PayPal	1,037	17.05%
UPI	1,010	16.61%
Total	**6,082**	**100%***

Individual percentages may not sum to exactly 100% because of rounding.

Interpretation: PayPal had the largest share, with 1,037 orders. Debit Card had the smallest share, with 985 orders. Shares ranged from 16.20% to 17.05%, indicating that no single method strongly dominated. These are order shares, not revenue shares.

Output File: payment_share.png

Insert the payment-method pie chart here.

4. Dashboard Mockup

A static dashboard combined the following elements on one page:

- Revenue, eligible orders, average order value, and units sold.
- Revenue comparison by category.
- Monthly revenue trend.
- Payment-method order shares.

The dashboard provides a compact overview of eligible sales performance.

Output File: dashboard_mockup.png

Insert the dashboard image here.

Tools Used

- **Python and Pandas:** KPI calculations, grouping, and analysis.
- **Matplotlib:** Charts and dashboard layout.
- **Jupyter Notebook:** Code, outputs, and documentation.

Deliverables

- Analysis notebook.
- KPI summary and written findings.
- Bar chart, line chart, and pie chart.
- Static dashboard mockup.
- Analysis report with recommendations and limitations.

5. Power BI Dashboard

An interactive Power BI dashboard was created to present sales KPIs and explore sales patterns. It includes revenue, order count, average order value, and units sold, along with visuals for category performance, monthly sales, order status, payment methods, and the top five countries by sales.

Sales visuals use eligible delivered records with positive quantities and available discounts. The order-status visual presents the statuses included in its own filter scope.

Dashboard File: ecommerce_dashboard.pbix

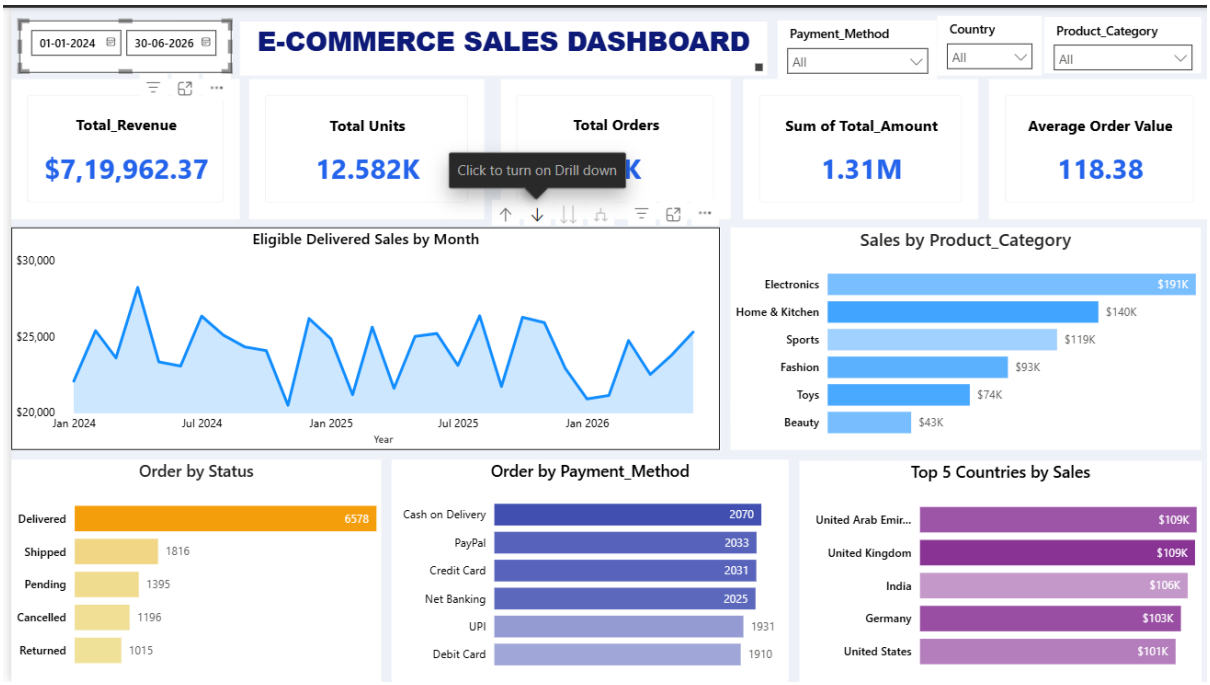


Fig 5 . Ecommerce Sales Dashboard in PowerBI